Module 6 Data Link Layer

6.1 Purpose of the Data Link Layer

The Data Link Layer

- Responsible for **communications between end-device network interface cards.**
- It allows upper layer protocols to access the physical layer media and encapsulates Layer 3 packets (IPv4 and IPv6) into Layer 2 Frames.
- It also performs error detection and rejects corrupted frames.

IEEE 802 LAN/MAN Data Link Sublayers

- IEEE 802 LAN/MAN standards are specific to the type of network (Ethernet, WLAN, WPAN, etc).

The Data Link Layer consists of two sublayers:

Logical Link Control (LLC)

Communicates between the networking software at the upper layers and the device hardware at the lower layers.

Media Access Control (MAC).

Responsible for data encapsulation and media access control.

Providing Access to Media

Packets exchanged between nodes may experience numerous data link layers and media transitions.

At each hop along the path, a router performs four basic Layer 2 functions:

- **Accepts a frame** from the network medium.
- **De-encapsulates** the frame to expose the encapsulated packet.
- **Re-encapsulates** the packet into a new frame.
- **Forwards the new frame** on the medium of the next network segment.

Data Link Layer Standards

Data link layer protocols are defined by engineering organizations:

- Institute for Electrical and Electronic Engineers **(IEEE)**.
- International Telecommunications Union **(ITU)**.
- International Organizations for Standardization **(ISO)**.
- American National Standards Institute **(ANSI)**.

6.2 Topologies

Physical and Logical Topologies

The topology of a network is the **arrangement and relationship of the network devices and the interconnections between them.**

There are two types of topologies used when describing networks:

- **Physical topology** – shows physical connections and how devices are interconnected.
- **Logical topology** – identifies the virtual connections between devices using device interfaces and IP addressing schemes.

WAN Topologies

There are three common physical WAN topologies:

Point-to-point – the simplest and most common WAN topology. Consists of a permanent link between two endpoints.

Hub and spoke – similar to a star topology where a central site interconnects branch sites through point-to-point links.

Mesh – provides high availability but requires every end system to be connected to every other end system.

Point-to-Point WAN Topology

Physical point-to-point topologies **directly connect two nodes**. The nodes may not share the media with other hosts. Because all frames on the media can only travel to or from the two nodes, Point-to-Point WAN protocols can be very simple.

LAN Topologies

End devices on LANs are typically **interconnected using a star or extended star topology**. Star and extended star topologies are easy to install, very scalable and easy to troubleshoot.

Early Ethernet and Legacy Token Ring technologies provide two additional topologies:

Bus – All **end systems chained together** and terminated on each end.
Ring – Each **end system is connected to its respective neighbors** to form a ring.

Half and Full Duplex Communication

Half-duplex communication
Only allows **one device to send or receive at a time** on a shared medium.
Used on WLANs and legacy bus topologies with Ethernet hubs.

Full-duplex communication
Allows both devices to simultaneously transmit and receive on a shared medium.
Ethernet switches operate in full-duplex mode.

Access Control Methods

Contention-based access
All nodes operating in half-duplex, **competing for use of the medium**. Examples are:

Carrier sense multiple access with collision detection (CSMA/CD) as **used on legacy bus-topology Ethernet.**

Carrier sense multiple access with collision avoidance (CSMA/CA) as **used on Wireless LANs.**

Controlled access
Deterministic access where **each node has its own time on the medium.**
Used on legacy networks such as Token Ring and ARCNET.

Contention-Based Access – CSMA/CD
CSMA/CD:
- Used by legacy Ethernet LANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CD collision detection process:
- Devices transmitting simultaneously will result in a signal collision on the shared media. Devices detect the collision. Devices wait a random period of time and retransmit data.

Contention-Based Access – CSMA/CA
CSMA/CA:
- Used by IEEE 802.11 WLANs.

- Operates in half-duplex mode where only one device sends or receives at a time.
Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CA collision avoidance process:
- When transmitting, devices also include the time duration needed for the transmission. Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

6.3 Data Link Frame

The Frame
Data is encapsulated by the data link layer with a header and a trailer to form a frame.
A data link frame has three parts:

- Header
- Data
- Trailer

The fields of the header and trailer vary according to data link layer protocol.
The amount of control information carried within the frame varies according to access control information and logical topology.

Frame Fields
- Layer 2 Addresses Also referred to as a physical address. Contained in the frame header. Used **only for local delivery of a frame** on the link. Updated by each device that forwards the frame.

LAN and WAN Frames
The logical topology and physical media determine the data link protocol used:

- Ethernet
- 802.11 Wireless
- Point-to-Point (PPP)
- High-Level Data Link Control (HDLC)
- Frame-Relay
- Each protocol performs media access control for specified logical topologies.

Ethernet Switching

Ethernet Frame
Explain how the Ethernet sublayers are related to the frame fields.

Ethernet MAC Address
Describe the Ethernet MAC address.

The MAC Address Table
Explain how a switch builds its MAC address table and forwards frames.

Switch Speeds and Forwarding Methods
Describe switch forwarding methods and port settings available on Layer 2 switch ports.

7.1 Ethernet Frames

Ethernet Encapsulation

- Ethernet operates in the data link layer and the physical layer.
- It is a family of networking technologies defined in the IEEE 802.2 and 802.3 standards.

Data Link Sublayers
The 802 LAN/MAN standards, including Ethernet, use two separate sublayers of the data link layer to operate:

- **LLC Sublayer: (IEEE 802.2)**
  Place information in the frame to identify which network layer protocol is used for the frame.
- **MAC Sublayer: (IEEE 802.3, 802.11, or 802.15)**

Responsible for data encapsulation and media access control, and provides data link layer addressing.

MAC Sublayer
The MAC sublayer is responsible for data encapsulation and accessing the media.

Data Encapsulation
IEEE 802.3 data encapsulation includes the following:
1. **Ethernet frame** - This is the internal structure of the Ethernet frame.
2. **Ethernet Addressing** - The Ethernet frame includes both a source and destination MAC address
to deliver the Ethernet frame from Ethernet NIC to Ethernet NIC on the same LAN.
3. **Ethernet Error detection** - The Ethernet frame includes a frame check sequence (FCS) trailer used for error detection.

Media Access

- The IEEE 802.3 MAC sublayer includes the specifications for different Ethernet communications standards over various types of media including copper and fiber.

- Legacy Ethernet using a bus topology or hubs, is a shared, half-duplex medium. Ethernet over a half-duplex medium uses a contention-based access method, carrier sense multiple access/collision detection (CSMA/CD).

- Ethernet LANs of today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.

Ethernet Frame Fields

- The **minimum Ethernet frame size is 64 bytes and the maximum is 1518 bytes.** The preamble field is not included when describing the size of the frame.

- Any frame **less than 64 bytes in length is considered a "collision fragment" or "runt frame"** and is automatically discarded. Frames with **more than 1500 bytes of data are considered "jumbo" or "baby giant frames".**

- If the size of a transmitted frame is less than the minimum, or greater than the maximum, the receiving device drops the frame. Dropped frames are likely to be the result of collisions or other unwanted signals. They are considered invalid. Jumbo frames are usually supported by most Fast Ethernet and Gigabit Ethernet switches and NICs.

7.2 Ethernet MAC Address

Ethernet MAC Address
• In an Ethernet LAN, every network device is connected to the same, shared media. MAC addressing provides a method for device identification at the data link layer of the OSI model.

• **An Ethernet MAC address is a 48-bit address expressed using 12 hexadecimal digits.** Because a byte equals 8 bits, we can also say that a MAC address is 6 bytes in length.

• **All MAC addresses must be unique to the Ethernet device or Ethernet interface**. To ensure this, all vendors that sell Ethernet

devices must register with the IEEE to obtain a unique 6 hexadecimal (i.e., 24-bit or 3-byte) code called the organizationally unique identifier (OUI).

• An Ethernet MAC address consists of a 6 hexadecimal vendor OUI code followed by a 6 hexadecimal vendor-assigned value.

Frame Processing

• When a device is forwarding a message to an Ethernet network, the Ethernet header includes a Source MAC address and a Destination MAC address.

• When a NIC receives an Ethernet frame, it examines the destination MAC address to see if it matches the physical MAC address that is stored in RAM. If there is no match, the device discards the frame. If there is a match, it passes the frame up the OSI layers, where the de-encapsulation process takes place.

Note: Ethernet NICs will also accept frames if the destination MAC address is a broadcast or a multicast group of which the host is a member.

• Any device that is the source or destination of an Ethernet frame, will have an Ethernet NIC and therefore, a MAC address. This includes workstations, servers, printers, mobile devices, and routers.

Unicast MAC Address

In Ethernet, different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.

• A unicast MAC address is the unique address that is used when a frame is sent from a single transmitting device to a single destination device.

• The process that a source host uses to determine the destination MAC address associated with an IPv4 address is known as Address Resolution Protocol (ARP). The process that a source host uses to determine the destination MAC address associated with an IPv6 address is known as Neighbor Discovery (ND).

Note: The source MAC address must always be a unicast.

Broadcast MAC Address

An Ethernet broadcast frame is received and processed by every device on the Ethernet LAN.

The features of an Ethernet broadcast are as follows:

• It has a destination MAC address of FF-FF-FF-FF-FF-FF in hexadecimal (48 ones in binary).

• It flooded out all Ethernet switch ports except the incoming port. It is not forwarded by a router.

• If the encapsulated data is an IPv4 broadcast packet, this means the packet contains a destination IPv4 address that has all ones (1s) in the host portion. This numbering in the address means that all hosts on that local network (broadcast domain) will receive and process the packet.

Multicast MAC Address

An Ethernet multicast frame is received and processed by a group of devices that belong to the same multicast group.

• There is a destination MAC address of 01-00-5E when the encapsulated data is an

IPv4 multicast packet and a destination MAC address of 33-33 when the encapsulated data is an IPv6 multicast packet.

• There are other reserved multicast destination MAC addresses for when the encapsulated data is not IP, such as Spanning Tree Protocol (STP).

• It flooded out all Ethernet switch ports except the incoming port, unless the switch is configured for multicast snooping. It is not forwarded by a router, unless the router is configured to route multicast packets.

• Because multicast addresses represent a group of addresses (sometimes called a host group), they can only be used as the destination of a packet. The source will always be a unicast address.

• As with the unicast and broadcast addresses, the multicast IP address requires a corresponding multicast MAC address.

7.3 The MAC Address Table

Switch Fundamentals

• A Layer 2 Ethernet switch uses Layer 2 MAC addresses to make forwarding decisions. It is completely unaware of the data (protocol) being carried in the data portion of the frame, such as an IPv4 packet, an ARP message, or an IPv6 ND packet. The switch makes its forwarding decisions based solely on the Layer 2

Ethernet MAC addresses

• An Ethernet switch examines its MAC address table to make a forwarding decision for each frame, unlike legacy Ethernet hubs that repeat bits out all ports except the incoming port.

• When a switch is turned on, the MAC address table is empty.

Note: The MAC address table is sometimes referred to as a content addressable memory (CAM) table.

Switch Learning and Forwarding

Examine the Source MAC Address (Learn)

Every frame that enters a switch is checked for new information to learn. It does this by examining the source MAC address of the frame and the port number where the frame entered the switch. If the source MAC address does not exist, it is added to the table along with the incoming port number. If the source MAC address does exist, the switch updates the refresh timer for that entry. By default, most Ethernet switches keep an entry in the table for 5 minutes.

Note: If the source MAC address does exist in the table but on a different port, the switch
treats this as a new entry. The entry is replaced using the same MAC address but with the
more current port number.

Switch Learning and Forwarding (Contd.)

Find the Destination MAC Address (Forward)

• If the destination MAC address is a unicast address, the switch will look for a match between the destination MAC address of the frame and an entry in its MAC address table.

• If the destination MAC address is in the table, it will forward the frame out the specified port.

• If the destination MAC address is not in the table, the switch will forward the frame out all ports except the incoming port. This is called an unknown unicast.

Note: If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

Filtering Frames

As a switch receives frames from different devices, it is able to populate its MAC address table by examining the source MAC address of every frame. When the MAC address table of the switch contains the destination MAC address, it is able to filter the frame and forward out a single port.

7.4 Switch Speeds and Forwarding Methods

Frame Forwarding Methods on Cisco Switches

Switches use one of the following forwarding methods for switching data between network ports:

• **Store-and-forward switching** - This frame forwarding method receives the entire frame and computes the CRC. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. Then the frame is forwarded out of the correct port.

• **Cut-through switching** - This frame forwarding method forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

• A big advantage of store-and-forward switching is that it determines if a frame has errors before propagating the frame. When an error is detected in a frame, the switch discards the frame. Discarding frames with errors reduces the amount of bandwidth consumed by corrupt data.

• Store-and-forward switching is required for **quality of service (QoS)** analysis on converged networks where frame classification for traffic prioritization is necessary. For example, **voice over IP (VoIP)** data streams need to have priority over web-browsing traffic.

Cut-Through Switching

In cut-through switching, the switch acts upon the data as soon as it is received, even if the transmission is not complete.

The switch buffers just enough of the frame to read the destination MAC address so that it can determine to which port it should forward out the data. The switch does not perform any error checking on the frame.

There are two variants of cut-through switching:

• **Fast-forward switching** - Offers the lowest level of latency by immediately forwarding a packet after reading the destination address. Because fast-forward switching starts forwarding before the entire packet has been received, there may be times when packets are relayed with errors. The destination NIC discards the faulty packet upon receipt. Fast- forward switching is the typical cut-through method of switching.

• **Fragment-free switching** - A compromise between the high latency and high integrity of store-and-forward switching and the low latency and reduced integrity of fast-forward switching, the switch stores and performs an error check on the first 64 bytes of the frame before forwarding. Because most network errors and collisions occur during the first 64 bytes, this ensures that a collision has not occurred before forwarding the frame.

Memory Buffering on Switches

An Ethernet switch may use a buffering technique to store frames before forwarding them or when the destination port is busy because of congestion.

• Shared memory buffering also results in larger frames that can be transmitted with fewer dropped frames. This is important with asymmetric switching which allows for different data rates on different ports. Therefore, more bandwidth can be dedicated to certain ports (e.g., server port).

Port-based memory

• Frames are stored in queues that are linked to specific incoming and outgoing ports.

• A frame is transmitted to the outgoing port only when all the frames ahead in the queue have been successfully transmitted.

• It is possible for a single frame to delay the transmission of all the frames in memory because of a busy destination port.

•This delay occurs even if the other frames could be transmitted to open destination ports.

Shared memory

•Deposits all frames into a common memory buffer shared by all switch ports and the amount of buffer memory required by a port is dynamically allocated.

•The frames in the buffer are dynamically linked to the destination port enabling a packet to be received on one port and then transmitted on another port, without moving it to a different queue.

Duplex and Speed Settings

Two of the most basic settings on a switch are the bandwidth (“speed”) and duplex settings for each individual switch port. It is critical that the duplex and bandwidth settings match between the switch port and the connected devices.

There are two types of duplex settings used for communications on an Ethernet network:

• **Full-duplex** - Both ends of the connection can send and receive simultaneously.

• **Half-duplex** - Only one end of the connection can send at a time.

Autonegotiation is an optional function found on most Ethernet switches and NICs. It enables two devices to automatically negotiate the best speed and duplex capabilities.

Note: Gigabit Ethernet ports only operate in full-duplex.

Duplex and Speed Settings

• **Duplex mismatch** is one of the most common causes of performance issues on 10/100 Mbps Ethernet links. It occurs when one port on the link operates at half- duplex while the other port operates at full-duplex.

• This can occur when one or both ports on a link are reset, and the autonegotiation

process does not result in both link partners having the same configuration.

• It also can occur when users reconfigure one side of a link and forget to reconfigure the other. Both sides of a link should have autonegotiation on, or both sides should have it off. Best practice is to configure both Ethernet switch ports as full-duplex.

Auto-MDIX

Connections between devices once required the use of either a crossover or straight- through cable. The type of cable required depended on the type of interconnecting devices.

Note: A direct connection between a router and a host requires a cross-over connection.

• Most switch devices now support the automatic medium-dependent interface crossover (auto-MDIX) feature. When enabled, the switch automatically detects the type of cable attached to the port and configures the interfaces accordingly.

• The auto-MDIX feature is enabled by default on switches running Cisco IOS Release 12.2(18)SE or later. However, the feature could be disabled. For this reason, you should always use the correct cable type and not rely on the auto-MDIX feature.

• Auto-MDIX can be re-enabled using the mdix auto interface configuration command.

Basic Router Configuration

Configure the device name.

Router(config)# hostname hostname

Secure privileged EXEC mode.

Router(config)# enable secret password

Secure user EXEC mode.

Router(config)# line console 0
Router(config-line)# password password
Router(config-line)# login

Secure remote Telnet / SSH access.

Router(config)# line vty 0 4
Router(config-line)# password password
Router(config-line)# login
Router(config-line)# transport input {ssh | telnet}

Encrypt all plaintext passwords.

Router(config)# service password encryption

Commands for basic router configuration on R1.

Router(config)# banner motd # message #
Router(config)# end
Router# copy running-config startup-config

```
R1(config)# hostname R1
R1(config)# enable secret
class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1( config-line)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config-line)#
exit
R1(config)# service password encryption
R1(config)# banner motd #
Enter TEXT message. End with a new line
and the #

***********************************************
WARNING: Unauthorized access
is
prohibited!
***********************************************
```

```
R1(config)# exit
R1# copy running-config startup-config
```

Configuration is saved to NVRAM.

10.2 Configure Interfaces

Configuring a router interface includes issuing the following commands:

Router(config)# interface type-and-number
Router(config-if)# description description-text
Router(config-if)# ip address ipv4-address subnet-mask
Router(config-if)# ipv6 address ipv6-address/prefix-length
Router(config-if)# no shutdown

description command
Used to add information about the network connected to the interface.

no shutdown command
activates the interface.

The commands to configure interface G0/0/0 on R1 are shown here:

```
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Aug 1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Aug 1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```

show ip interface brief **show ipv6 interface brief**	Displays all interfaces, their IP addresses, and their current status.
show ip route **show ipv6 route**	Displays the contents of the IP routing tables stored in RAM.
show interfaces	Displays statistics for all interfaces on the device. Only displays the IPv4 addressing information.
show ip interfaces	Displays the IPv4 statistics for all interfaces on a router.
show ipv6 interfaces	Displays the IPv6 statistics for all interfaces on a router.

show ip route and **show ipv6 route commands**
Used to display the contents of the IP routing tables

show interfaces command
Display statistics for all interfaces

show ip interface command
Display IPv4 statistics for router interfaces
show ipv6 interface command
Display IPv6 statistics for router interfaces

10.3 Configure the Default Gateway

Default Gateway on a Host

• The default gateway is used when a host sends a packet to a device on another network.

• The default gateway address is generally the router interface address attached to the local network of the host.

• To reach PC3, PC1 addresses a packet with the IPv4 address of PC3, but forwards the packet to its default gateway, the G0/0/0 interface of R1.

Note: The IP address of the host and the router interface must be in the same network.

Default Gateway on a Switch

• A switch must have a default gateway address configured to remotely manage the switch from another network.

• To configure an IPv4 default gateway on a switch, use the ip default-gateway ip-address global configuration command.